

## Chapter 1: Functions and Graphs

MATH 1334

September 25, 2026

These notes follow the Chapter 1 review worksheet in order. Each numbered topic corresponds to the same numbered worksheet problem. The examples use different values and are intended for discussion.

### Functions and representations

#### 1. Functions, domain, and range

A *function* assigns exactly one output to each input in its domain. The *domain* is the set of allowed inputs; the *range* is the set of outputs the function actually produces. The notation  $f(x)$  means the output corresponding to input  $x$ . A point on its graph has coordinates  $(x, f(x))$ .

A function may be represented by a verbal description, table, formula, or graph. For a quantity changing at a constant rate,

$$\text{amount} = \text{initial amount} + (\text{rate})(\text{elapsed time}).$$

Identify the input and output, label their units, and use the situation to decide which inputs are allowed. A formula alone may allow more inputs than the situation does.

*Example.* A tank contains 6 liters of water. Water enters at 3 liters per minute for 4 minutes. Let  $W(t)$  be the volume after  $t$  minutes.

- (a) Make a table for  $t = 0, 1, 2, 3, 4$  and write a formula for  $W(t)$ .
- (b) Sketch the graph with labeled axes and units. Find  $W(2)$  and interpret it.
- (c) State the domain and range for this situation.

#### 2. Evaluating functions and simplifying a difference quotient

To evaluate a function, replace every occurrence of its input variable by the given number or expression. Use parentheses around the entire substituted input. In general,  $f(a + h)$  is not equal to  $f(a) + f(h)$ .

A *difference quotient* has the form

$$\frac{f(a + h) - f(a)}{h}, \quad h \neq 0.$$

Substitute first, expand, subtract all of  $f(a)$ , and then collect terms and factor. Remember  $(a + h)^2 = a^2 + 2ah + h^2$ . Cancel common *factors*, not individual terms in a sum, and retain the restriction  $h \neq 0$ .

*Example.* Let  $f(x) = x^2 - 4x$ . Find  $f(-3)$  and expand  $f(a + h)$ . Then simplify  $[f(a + h) - f(a)]/h$  for  $h \neq 0$ , showing the factoring and cancellation step.

#### 3. Composition of functions

The *composition*  $(g \circ h)(x) = g(h(x))$  applies  $h$  first and then applies  $g$  to the result. Read from the inside out. The order generally matters.

*Example.* Let  $g(x) = x + 3$  and  $h(x) = x^2$ . Find  $g(h(x))$  and  $h(g(x))$ . Describe the order of operations in each.

*Review:* OpenStax, *Calculus Volume 1*, §1.1; §1.2 for linear functions.

## Graphs and domain

### 4. Piecewise-defined functions

A *piecewise-defined function* uses different formulas on different parts of its domain. To evaluate it, first determine which condition the input satisfies; then use the corresponding rule.

To graph it, draw each formula only for the inputs covered by its condition. An open endpoint means that branch excludes the point; a filled endpoint means the point belongs to the graph. An equality condition may specify a single point. When branches meet, check whether any rule includes that point in the whole function.

*Example.* Consider

$$r(x) = \begin{cases} 2x + 1, & x < 2, \\ 4, & x = 2, \\ 6 - x, & x > 2. \end{cases}$$

- (a) Find  $r(1)$ ,  $r(2)$ , and  $r(3)$ . Identify the rule used for each input.
- (b) Sketch the graph with open and filled endpoints clearly marked.
- (c) Which point is included at  $x = 2$ ? Explain how you know.

### 5. Finding real domains

When no domain is stated, seek all real inputs for which the formula gives a defined real output. The main restrictions are

- a fraction: denominator  $\neq 0$ ,
- a square root: radicand  $\geq 0$ ,
- a logarithm: argument  $> 0$ .

If a formula has more than one restriction, satisfy all of them simultaneously.

In interval notation, brackets include a finite endpoint and parentheses exclude it. Always use parentheses with  $\infty$  and  $-\infty$ . The union symbol  $\cup$  joins separate intervals. For example,  $(a, b]$  means  $a < x \leq b$ .

*Example.* Find each real domain. Explain the restriction before writing interval notation.

$$(a) u(x) = \sqrt{x - 3}, \quad (b) v(x) = \frac{1}{x^2 - 16}, \quad (c) w(x) = \ln(x + 2).$$

### 6. Simplification and excluded inputs

The domain is determined from the original expression. Canceling a common factor does not restore an input excluded by that expression. The identity

$$x^2 - c^2 = (x - c)(x + c)$$

is useful for factoring a difference of squares. After cancellation, keep the original restrictions alongside the simplified formula.

If a simplified formula is defined at an originally excluded input, its graph must have that point removed to represent the original function. Mark the missing point with an open circle; use the simplified formula to locate its height.

*Example.* A student claims that  $(x^2 - 9)/(x - 3) = x + 3$  for every real  $x$ . Correct the statement, sketch the graph of the original function, and compare its domain with the domain of  $y = x + 3$ .

*Review:* OpenStax, *Calculus Volume 1*, §1.1; §1.2 for piecewise and rational functions; §1.5 for logarithms.

## Slope and trigonometry

### 7. Slope, average rate of change, and lines

The slope of the line through two points  $(x_1, y_1)$  and  $(x_2, y_2)$  is

$$m = \frac{y_2 - y_1}{x_2 - x_1}, \quad x_1 \neq x_2.$$

Subtract in the same order in the numerator and denominator. Slope has units of output units per input unit. For points on  $y = f(x)$ , this is the *average rate of change* over the interval between the inputs.

A *secant line* passes through two distinct points on a graph. Use point-slope form to write its equation:

$$y - y_1 = m(x - x_1).$$

For a position function  $s(t)$ , the slope between two points is average velocity. The secant line need not coincide with the position graph between those points.

*Example.* An object's position is  $s(t) = t^2 + 1$  meters, with  $t$  in seconds.

- (a) Find the slope between the points at  $t = 2$  and  $t = 3$ . Interpret it with units.
- (b) Repeat for  $t = 2$  and  $t = 2.5$ . Compare the two slopes.
- (c) Write the equation of the line from part (a), using  $t$  and  $s$  as coordinates.

### 8. Radians and the unit circle

An angle in standard position is measured from the positive horizontal axis; positive angles turn counterclockwise. A full turn is  $2\pi$  radians, so

$$\pi \text{ radians} = 180^\circ, \quad \text{degrees} = \text{radians} \cdot \frac{180^\circ}{\pi}.$$

The point at angle  $\theta$  on the unit circle has coordinates

$$(\cos \theta, \sin \theta).$$

Cosine is the horizontal coordinate and sine is the vertical coordinate. Use the quadrant to determine signs. The special right triangles give exact values: a  $30^\circ$ – $60^\circ$ – $90^\circ$  triangle has side ratios  $1 : \sqrt{3} : 2$ ; a  $45^\circ$ – $45^\circ$ – $90^\circ$  triangle has ratios  $1 : 1 : \sqrt{2}$ .

*Example.* Convert  $\pi/3$ ,  $\pi/4$ , and  $3\pi/2$  to degrees and give their unit-circle coordinates. Identify the sine and cosine in each pair. A unit-circle reference may be used.

### 9. Sine graphs, amplitude, and period

The graph of  $y = \sin x$  records the vertical coordinate on the unit circle. The *amplitude* is the distance from the midline to a peak. The *period* is the length of one complete repeating cycle.

For  $y = A \sin x$  with  $A \neq 0$ , the amplitude is  $|A|$  and the period is  $2\pi$ . Multiplying the output changes vertical heights; it does not change the input needed for one cycle. To sketch a cycle, locate the values at  $0, \pi/2, \pi, 3\pi/2, 2\pi$  and join them with a smooth curve.

*Example.* On the same axes, sketch  $y = \sin x$  and  $y = 3 \sin x$  for  $0 \leq x \leq 2\pi$ . Label the intercepts, maxima, and minima, and state each graph's amplitude and period.

*Review:* OpenStax, *Calculus Volume 1*, §1.2 for slope, lines, and scaling; §1.3 for trigonometry.

## Inverses, exponentials, and logarithms

### 10. Inverse functions

An *inverse function* reverses the input-output relationship: if  $f(a) = b$ , then  $f^{-1}(b) = a$ . Its domain is the range of  $f$ , and its range is the domain of  $f$ . On the appropriate domains,

$$f^{-1}(f(x)) = x, \quad f(f^{-1}(x)) = x.$$

To find an inverse formula, write  $y = f(x)$ , solve for  $x$  in terms of  $y$ , and then use  $x$  as the input variable in the inverse formula. Equivalently, interchange  $x$  and  $y$  and solve for  $y$ . Check the result by composition.

The superscript in  $f^{-1}$  denotes an inverse function, not a reciprocal. The expression  $1/f(x)$  means the reciprocal of the output and requires  $f(x) \neq 0$ .

*Example.* Let  $f(x) = 4x - 8$ . Find  $f^{-1}(x)$  and verify it by simplifying  $f(f^{-1}(x))$ . Write  $1/f(x)$  as well, and explain how it differs from the inverse.

### 11. One-to-one functions and domain restrictions

A function is *one-to-one* if different inputs always have different outputs. It has an inverse function on its range exactly when it is one-to-one. Graphically, this means that every horizontal line intersects the graph at most once (the *horizontal line test*).

If a function is not one-to-one, a suitable restriction of its domain may make it so. Choose a portion of the graph on which no output repeats. The inverse must return an input in that chosen domain; its own domain is the range of the restricted original function. Recall that  $\sqrt{x}$  denotes the nonnegative square root.

*Example.* Consider  $q(x) = x^2 + 1$  on all real numbers.

- Find two different inputs with the same output. Explain why there is no inverse function on the stated domain.
- Restrict the domain so that an inverse function exists. Find the inverse and state its domain.
- What changes if you choose the other side of the parabola?

### 12. Exponentials and logarithms

A logarithm answers the question “What exponent produces this number?” For  $b > 0$ ,  $b \neq 1$ , and  $v > 0$ ,

$$b^u = v \iff \log_b v = u.$$

The *natural logarithm* is  $\ln x = \log_e x$ . Exponential and logarithmic functions with the same base undo each other:

$$\log_b(b^u) = u \quad (u \in \mathbb{R}), \quad b^{\log_b v} = v \quad (v > 0).$$

To solve an exponential equation, apply a logarithm to both sides. For positive  $a, b$  and real  $r$ , the natural-logarithm rules include

$$\ln(ab) = \ln a + \ln b, \quad \ln(a/b) = \ln a - \ln b, \quad \ln(a^r) = r \ln a.$$

The product rule applies to multiplication inside the logarithm. It does not give a rule for splitting the logarithm of a sum. One counterexample disproves a claim that an identity always holds.

*Example.* Work without a calculator.

- Rewrite  $3^4 = 81$  in logarithmic form. Evaluate  $\log_2 8$  and  $\ln(e^5)$ .
- Solve  $e^x = 11$  exactly.
- Test the claim  $\ln(a + b) = \ln a + \ln b$  using positive values of your choice.

*Review:* OpenStax, *Calculus Volume 1*, §1.4 for inverses; §1.5 for exponentials and logarithms.